

AI-Powered Customer Support & Ticket Management System

Comprehensive Project Report & Architecture Documentation

1. Executive Summary

This project presents a modern, production-ready AI Customer Support platform built using Python 3.12, Flask, Flask-SQLAlchemy, Flask-Login, and Bootstrap 5. It delivers instant unauthenticated guest AI support alongside automated ticket escalation.

2. Student & Project Metadata

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Department: Computer Science & Engineering (CSE)

College: Sree Dattha Group of Institutions

Guide: Dr. A. Ramesh Kumar (HOD & Professor)

3. System Architecture & Objectives

- Instant guest AI chat assistance without customer login barriers.
- Pre-LLM Knowledge Base RAG lookup reducing operational API costs.
- Automated ticket escalation generating unique Ticket IDs (TICK-XXXXXXX).
- Centralized Admin Console with Chart.js analytics and CSV exporter.